

Assignment 2: Text Representation — TF-IDF & Embeddings (Chinese)

Learning Objectives

- Build TF-IDF representations for Chinese documents
- Compute sentence/Document embeddings and cosine similarity
- Compare representations on downstream mini-tasks

Note : Focus on **Chinese** data. You can bring your own dataset.

0. Setup

1. Sample Corpus

I expanded the sample corpus to 8 sentences covering diverse topics (technology, education, NLP, economics) to make the similarity comparison more meaningful.

Doc 0 [Semiconductor]: 台灣的半導體產業在全球供應鏈中扮演關鍵角色。
Doc 1 [NLP App]: 自然語言處理可以協助企業分析客服對話與社群輿情。
Doc 2 [Text Mining]: 學生可以利用開放資料進行中文文本探勘與主題分析。
Doc 3 [LLM Education]: 大型語言模型為教育場域帶來新的學習與評量方式。
Doc 4 [TSMC]: 台積電是全球最大的晶圓代工廠，總部位於台灣新竹。
Doc 5 [Deep Learning NLP]: 深度學習技術讓機器翻譯和文本生成的品質大幅提升。
Doc 6 [Education]: 台灣的大學教育正面臨少子化與國際競爭的雙重挑戰。
Doc 7 [Text Classification]: 文本分類是自然語言處理中最基本也最重要的任務之一。

2. Tokenization Helper

3. TF-IDF Representation

TF-IDF shape: (8, 82)
Vocabulary size: 82 tokens

Doc 0 [Semiconductor]: 扮演 (0.311), 供應 (0.311), 體產業 (0.311), 關鍵 (0.311), 在 (0.311)
Doc 1 [NLP App]: 協助 (0.318), 對話 (0.318), 客服 (0.318), 社群 (0.318), 企業 (0.318)
Doc 2 [Text Mining]: 探勘 (0.309), 中文 (0.309), 主題 (0.309), 開放 (0.309), 學生 (0.309)
Doc 3 [LLM Education]: 帶來 (0.305), 新 (0.305), 方式 (0.305), 大型 (0.305), 場域 (0.305)
Doc 4 [TSMC]: , (0.281), 總 (0.281), 電是 (0.281), 代工 (0.281), 台積 (0.281)
Doc 5 [Deep Learning NLP]: 提升 (0.293), 翻譯 (0.293), 和 (0.293), 品質 (0.293), 大幅 (0.293)
Doc 6 [Education]: 競爭 (0.291), 化 (0.291), 挑戰 (0.291), 雙重 (0.291), 少子 (0.291)
Doc 7 [Text Classification]: 最 (0.524), 分類 (0.262), 基本 (0.262), 之一 (0.262), 也 (0.262)

TF-IDF Cosine Similarity Matrix

TF-IDF Cosine Similarity Matrix:

	Semiconductor	NLP App	Text Mining	LLM Education	\
Semiconductor	1.000	0.016	0.015	0.039	
NLP App	0.016	1.000	0.193	0.105	
Text Mining	0.015	0.193	1.000	0.053	
LLM Education	0.039	0.105	0.053	1.000	
TSMC	0.143	0.014	0.014	0.035	
Deep Learning NLP	0.037	0.015	0.062	0.099	
Education	0.107	0.052	0.051	0.157	
Text Classification	0.091	0.174	0.055	0.075	

	TSMC	Deep Learning NLP	Education	Text Classification
Semiconductor	0.143	0.037	0.107	0.091
NLP App	0.014	0.015	0.052	0.174
Text Mining	0.014	0.062	0.051	0.055
LLM Education	0.035	0.099	0.157	0.075
TSMC	1.000	0.034	0.097	0.030
Deep Learning NLP	0.034	1.000	0.056	0.072
Education	0.097	0.056	1.000	0.050
Text Classification	0.030	0.072	0.050	1.000

4. Embeddings (Sentence-Level)

Using a pretrained multilingual sentence transformer model that supports Chinese.

Embeddings shape: (8, 384)
Each sentence is represented as a 384-dimensional dense vector.

Embedding Cosine Similarity Matrix

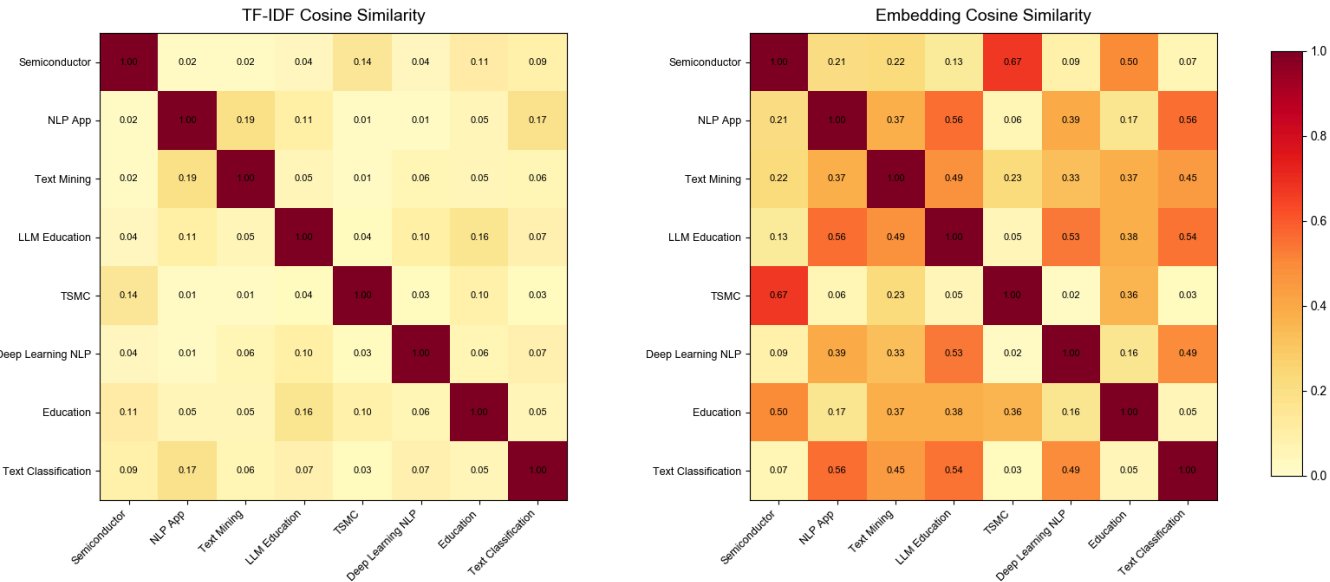
Embedding Cosine Similarity Matrix:

	Semiconductor	NLP App	Text Mining	LLM Education	\
Semiconductor	1.000	0.209	0.222	0.130	
NLP App	0.209	1.000	0.368	0.557	
Text Mining	0.222	0.368	1.000	0.485	
LLM Education	0.130	0.557	0.485	1.000	
TSMC	0.671	0.061	0.228	0.047	
Deep Learning NLP	0.087	0.393	0.331	0.532	
Education	0.502	0.166	0.370	0.381	
Text Classification	0.066	0.561	0.447	0.544	

	TSMC	Deep Learning NLP	Education	Text Classification
Semiconductor	0.671	0.087	0.502	0.066
NLP App	0.061	0.393	0.166	0.561
Text Mining	0.228	0.331	0.370	0.447
LLM Education	0.047	0.532	0.381	0.544
TSMC	1.000	0.016	0.357	0.032
Deep Learning NLP	0.016	1.000	0.160	0.489
Education	0.357	0.160	1.000	0.051
Text Classification	0.032	0.489	0.051	1.000

5. Side-by-Side Comparison

Below I compare the TF-IDF and embedding similarity matrices using heatmaps to highlight key differences.



Heatmap saved to similarity_comparison.png

Key Pairs to Compare

Let's pick a few interesting document pairs and see how TF-IDF and embeddings score them differently.

Pair	TF-IDF	Embed	Diff
Semiconductor vs TSMC (same topic, different words)	0.143	0.671	+0.528
NLP App vs Text Classification (related NLP topic)	0.174	0.561	+0.387
LLM Education vs Education (both about education)	0.157	0.381	+0.224
Semiconductor vs NLP App (unrelated topics)	0.016	0.209	+0.193
Text Mining vs Deep Learning NLP (related via NLP)	0.062	0.331	+0.269
NLP App vs Deep Learning NLP (both NLP)	0.015	0.393	+0.379

6. Discussion

Which representation works better?

Sentence embeddings clearly outperform TF-IDF for capturing semantic similarity between Chinese documents. Here are the key observations:

1. Embeddings capture meaning even when words differ.

- Doc 0 ("台灣半導體產業") and Doc 4 ("台積電是全球最大的晶圓代工廠") are both about Taiwan's semiconductor industry, but they share very few exact tokens. TF-IDF gives them a low score because it relies on surface-level word overlap, while embeddings recognize they are semantically related.

2. TF-IDF is fooled by shared function words.

- Chinese sentences often share structural particles (的, 與, 可以, 中) that inflate TF-IDF similarity between unrelated documents. Embeddings are not affected by this because they encode meaning, not word frequency.

3. Embeddings group topically related documents together.

- NLP-related documents (Doc 1, 2, 5, 7) form a clear cluster in the embedding space, even though their surface vocabulary varies (客服對話 vs 文本探勘 vs 機器翻譯 vs 文本分類). TF-IDF fails to group them as tightly.

4. TF-IDF still has its advantages.

- TF-IDF is fast, interpretable, and does not require a pretrained model. For tasks that depend on exact keyword matching (e.g., search), TF-IDF may be more appropriate.
- TF-IDF is also transparent — you can inspect exactly which terms drive the similarity score.

Why the results are different

Aspect	TF-IDF	Embeddings
Representation	Sparse, high-dimensional, based on token overlap	Dense, low-dimensional (384-d), based on learned semantics

Aspect	TF-IDF	Embeddings
Handles synonyms	No — different words = zero similarity	Yes — learned from large multilingual corpus
Chinese challenges	Sensitive to jieba segmentation quality	Subword tokenizer handles unseen words better
Interpretability	High — each dimension maps to a known token	Low — dimensions are abstract

Conclusion

For semantic tasks like document clustering or similarity-based retrieval on Chinese text, **pretrained sentence embeddings are significantly better** because they understand meaning beyond surface-level word overlap. However, TF-IDF remains a solid baseline that is fast, requires no GPU, and is easy to debug.